

Unimodularity

LP vs ILP

Q: Under which circumstances does optimal solution take integer values?

Def: A $p \times q$ integer matrix $rank(A) = p$

A is unimodular iff: for any basis B of A , $det(B) = \pm 1$

Thm The following are equivalent:

- i. A unimodular
- ii. Basic solution of $Ax = b$ (b is integer) are integer
- iii. B^{-1} integer $\forall$ basis B of A

Proof A integer matrix, B submatrix of A so B integer

Start with i. $\rightarrow$ ii.:

A unimodular $\rightarrow$ basic solutions

Solution solves $Bx = b$ given by Cramer's rule

$$x_i = \frac{\det(B \text{ where } i\text{th column replaced by } b)}{\det(B)}$$

All components are integer with the denominator is ± 1 and calculating a determinant is addition, subtraction, multiplication but no division. Thus determinant in the numerator is integer.

Now ii. $\rightarrow$ iii.:

e_j : j th component, $B^{-1}e_j$: j th component of B^{-1} , d_i : i th component of $B^{-1}e_j$

$$\alpha_i = \begin{cases} 0 & d_i \geq 0 \\ \lceil -d_i \rceil & d_i < 0 \end{cases}$$

Thus helps us get the following notes: α_i integer, $\alpha_i \geq 0$, $\alpha_i + d_i \geq 0$

Solve: $Bx = e_j + B\alpha$, if we take inverse of B of both sides (note that we are choosing $e_j + B\alpha$ as our b)

$\rightarrow x = B^{-1}e_j + \alpha$ since x, α are integer then d_i must also be integer thus B^{-1} is integer

Lastly iii. $\rightarrow$ i.

B^{-1} integer

Which means that $1 = \det(B)\det(B^{-1})$ thus there are only two ways this statement holds and that is $\det(B) = \det(B^{-1}) = \pm 1$ meaning A is unimodular.